

Doug Suttles, Ookla: India's National Broadband Mission highlights satellite as part of the technology mix to extend broadband connectivity across India. Satellite internet services will serve as a competitive play in areas where fiber hasn't been deployed, extending coverage to remote places. Our recent analysis shows that Starlink provides a much-loved option for more rural, nonmetro users who often don't have many good internet options in the US. Starlink users are more than willing to recommend the service and love the internet they're getting. There's broader roles for satellite internet, both in terms of providing backhaul services and offering services to enterprise verticals.

Q Every second, how many speed requests are processed by Ookla globally?

A Over 18 million tests are taken with Speedtest daily, with more than 45 billion tests taken to date.

Q Can you give us an insight into the data flowing in and flowing at the edge locations? Is it processed locally at the edge, and where do the speed test servers sit (where are they located)?

A Ookla's Speedtest Server Network has over 18,000 servers distributed across the world. This means that most users test their performance using a server that's located near to them. This is important, as it allows Speedtest to measure the performance of the connection itself (Quality of Service) in addition to the performance that user experiences with the rest of the internet (Quality of Experience).

Q With 5G becoming more prevalent in India, has Ookla seen a significant uptick in internet speeds? Can you provide some statistics?

A Yes, we've seen India climbing quite a few ranks on the Speedtest Global Index following the launch of 5G. Speedtest Intelligence data shows

that median mobile download speeds across India increased significantly since the 5G launch, from a 13.87 Mbps median download speed in September 2022 to 39.94 Mbps in May 2023. As a result, India's global rank for overall median mobile download speeds on the Speedtest Global Index has improved, moving from 118th in September 2022 to 56th in May 2023.

Q When do you think satellite communication services will become more commercially available?

A That very much depends on the outcome of TRAI's consultation on whether or not to award spectrum,



or allocate it to Satcom players. Recently, TRAI has further extended the deadlines for submission of counter comments. For example, OneWeb, backed by the Bharti Group, hopes to be able to launch services in late September 2023.

Q We heard Ookla may be planning to publish rankings for video and gaming experiences. Can you elaborate?

A At Ookla, we're focused on measuring, understanding, and helping improve connected experiences globally. This includes video and gaming experiences. While we don't publish these rankings currently, we're constantly working behind the scenes to create new publicly available resources that are in line with our methodologies and provide data and analyses that are meaningful, accurate, and statistically sound. With time, we'll be able to start providing these rankings publicly. Ookla also measures latency in several different ways, which is important as well when it comes to video and gaming experiences.

Q 5G hasn't been a drastic increase in speed like we once expected. What's the reason? Will it increase once the technology becomes more commonplace in India?

A Many marketing messages have promised Gigabit speeds for 5G, but the actual speeds depend heavily on multiple factors such as deployment type, spectrum holdings, and backhaul infrastructure. In India, comparing Speedtest Intelligence data from May 2023, we already saw significant improvement. The median 5G download speed is at least 20 times faster than 4G LTE, with a speed of

322.85 Mbps versus 15.87 Mbps, and the median 5G upload speed is also at least 5 times faster than 4G LTE, with a speed of 19.64 Mbps versus 3.58 Mbps. However,

the median 5G performance only represents the average user experience, and there is a significant variance in consumer experience on today's 5G networks.

In India, for example, according to Speedtest Intelligence, 5G speeds range from 662.15 Mbps for the best 10% to 31.80 Mbps for the worst 10%. This is similar to well-established 5G markets like South Korea, where the median 5G download speed was 510.68 Mbps in May 2023, with speeds ranging from close to 1 Gbps for the top 10% (982.83 Mbps) to 137.29 Mbps for the worst 10%.

Q What does the name "Ookla" mean? How did it come around?

A When Ookla was established back in 2006, I suggested that name for the company. I already owned the Ookla.com domain name to honor my cat, who was named after a character on the TV series, "Thundarr the Barbarian." If you look back at what existed on Ookla.com before 2006, you will find a picture of Ookla as a kitten curled up in a sink! 